AP BIOLOGY TOPIC 6.5

Regulation of Gene Expression: Epigenetics & Bacterial Operons

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-2.A | IST-2.B | LAC VS. TRP OPERON | EPIGENETICS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.5 (REGULATION OF GENE EXPRESSION: EPIGENETICS & BACTERIAL OPERONS)

CED Core Vocabulary

Purpose of Gene Regulation — Conserves cellular energy/resources in prokaryotes and allows multicellular eukaryotes with the exact same genome to differentiate into distinct cell types by...

Operator (Prokaryotes) — DNA switch segment within/near the promoter that binds a repressor protein to block RNA Polymerase.

Activators (Positive Control) — Bind enhancers/promoters to recruit RNA Polymerase → turn transcription ON / UP .

Epigenetics — Heritable, reversible chemical modifications to DNA or histone proteins that alter chromatin packing and gene accessibility without altering the DNA base sequence .

Promoter — DNA sequence upstream of a gene where RNA Polymerase and general transcription factors bind to initiate transcription.

Enhancers & Silencers (Eukaryotes) — Distal control regions (often thousands of bp away) that bind activators (boost transcription) or repressors (decrease transcription).

Repressors (Negative Control) — Bind operators/silencers to block RNA Polymerase → turn transcription OFF / DOWN .

Coordinated Eukaryotic Regulation — Eukaryotes do NOT have operons; instead, genes in the same pathway located on different chromosomes share the same combination of regulatory control elements so...

§2. WHY REGULATE GENES? REGULATORY SEQUENCES & PROTEINS

Right Time, Place, Amount

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Why Regulate Genes? Regulatory Sequences & Proteins ➔

Purpose of Gene Regulation ➔ **Regulatory Proteins (trans-acting factors)**

- **Purpose of Gene Regulation:** Conserves cellular energy/resources in prokaryotes and allows multicellular eukaryotes with the **exact same genome** to differentiate into distinct cell types by expressing proteins at the **right time, right place, and right amount**.
- **Regulatory DNA Sequences (*cis*-acting elements on DNA):**
  - **Promoter:** DNA sequence upstream of a gene where **RNA Polymerase** and general transcription factors bind to initiate transcription.
  - **Operator (Prokaryotes):** DNA switch segment within/near the promoter that binds a **repressor protein** to block RNA Polymerase.
  - **Enhancers & Silencers (Eukaryotes):** Distal control regions (often thousands of bp away) that bind **activators** (boost transcription) or **repressors** (decrease transcription).
- **Regulatory Proteins (*trans*-acting factors):**
  - **Activators (Positive Control):** Bind enhancers/promoters to recruit RNA Polymerase → turn transcription **ON / UP**.
  - **Repressors (Negative Control):** Bind operators/silencers to block RNA Polymerase → turn transcription **OFF / DOWN**.

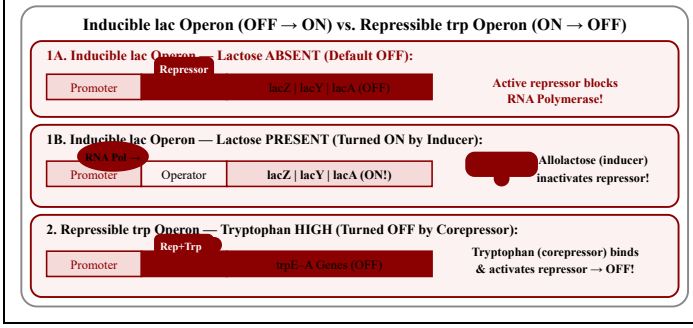
[CORE AXIOM / REMEMBER] **Activators = ON (Gas Pedal) • Repressors = OFF (Brake Pedal).**

§5. PROKARYOTIC OPERONS: INDUCIBLE (LAC) VS. REPRESSIBLE (TRP)

Master Bacterial Control

- **What Is an Operon?** A bacterial cluster of functionally related **structural genes** transcribed together as a single polycistronic mRNA under the control of **1 Promoter** and **1 Operator** (plus an upstream *regulatory gene* encoding the repressor protein).

FEATURE	INDUCIBLE OPERON (E.G., *LAC* OPERON)	REPRESSIBLE OPERON (E.G., *TRP* OPERON)
Default State	**Normally OFF** (Repressor synthesized in **active** shape bound to Operator)	**Normally ON** (Repressor synthesized in **inactive** shape; cannot bind alone)
Metabolic Pathway	**Catabolic (Breakdown):** Makes enzymes (β-galactosidase) ONLY when nutrient is present	**Anabolic (Synthesis):** Makes enzymes to build amino acid UNTIL product accumulates
Effector Molecule Role	**Inducer (Allolactose / Lactose):** Binds repressor → allosterically **inactivates repressor** → repressor falls off operator → **Turns Operon ON**!	**Corepressor (Tryptophan):** Binds inactive repressor → allosterically **activates repressor** to bind operator → **Turns Operon OFF**!
Positive Control (**cAMP–CAP** **)**	When **Glucose is LOW**, high **cAMP** binds **CAP activator** at promoter → **HIGH transcription** (only when Lactose PRESENT + Glucose ABSENT).	Regulated by negative corepressor feedback (and ribosome attenuation).



[AP EXAM FRQ SCORING STRATEGY] **Classic AP Operon Mutation Scenarios:**

- **Mutated Operator (cannot bind repressor) OR Mutated Regulatory Gene (** **$lacI^-$** **):** Repressor can never bind → operon is **constitutively ON (always transcribed)**!
- **Mutated Promoter:** RNA Polymerase can never bind → operon is **permanently OFF**!

§3. EPIGENETIC CHROMATIN REGULATION (EUCHROMATIN VS. HETEROCHROMATIN)

No DNA Sequence Change

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Epigenetic Chromatin Regulation (Euchromatin vs. Heterochromatin) ➔

Epigenetics ➔ **1. Histone Acetylation (-$COCH_3$)**

- **Epigenetics:** Heritable, reversible chemical modifications to **DNA or histone proteins** that alter chromatin packing and gene accessibility **without altering the DNA base sequence**.
- **1. Histone Acetylation (-$COCH_3$):**
  - Neutralizes positive charges on histone Lysine tails → weakens grip on negatively charged DNA → loosens chromatin into open **Euchromatin** → **INCREASES transcription (Gene ON)**.
- **2. DNA Methylation (-CH_3 on Cytosine bases):**
  - Adds methyl groups directly to promoter cytosines (CpG islands) and recruits deacetylases → condenses chromatin into tightly packed **Heterochromatin** → **SILENCES transcription (Gene OFF)** (e.g., X-chromosome Barr body inactivation, genomic imprinting).

CHROMATIN STATE	PACKING & CHEMICAL TAGS	TRANSCRIPTION LEVEL
Euchromatin	**Loose / Open** (High **Histone Acetylation**, Low DNA Methylation)	**HIGH (Active ON)** — RNA Polymerase can access promoters
Heterochromatin	**Tight / Condensed** (High **DNA Methylation**, Histone Deacetylation)	**LOW / OFF (Silenced)** — Promoters physically blocked

[CORE AXIOM / REMEMBER] **Memory Hook:** Histone **A**cetylation = **A**ctive / **A**ccessible (Euchromatin ON); DNA **M**ethylation = **M**uted (Heterochromatin OFF)!

§4. EUKARYOTIC MULTI-LEVEL & POST-TRANSLATIONAL CONTROL

No Operons in Eukaryotes

- **Coordinated Eukaryotic Regulation:** Eukaryotes do **NOT** have operons; instead, genes in the same pathway located on different chromosomes share the **same combination of regulatory control elements** so a single transcription factor activates all of them simultaneously.
- **Post-Translational Modifications (After Protein Synthesis):**
  - **Phosphorylation:** Kinases add -PO_4^{3-} (phosphatases remove it) to rapidly switch enzyme activity ON or OFF.
  - **Ubiquitination:** Attaches *ubiquitin* tags to damaged or unneeded proteins, targeting them for destruction inside the **proteasome**.
  - **Glycosylation / Cleavage:** Adds carbohydrate tags for membrane targeting or cleaves pro-proteins (e.g., pro-insulin → active insulin).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 6.5: Regulation of Gene Expression: Epigenetics & Bacterial Operons ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.5 to prove why the claim holds.

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